

Next, Exhibit 3.6 attempts to give a clearer picture of how sequence-of-returns risk impacts both the accumulation and distribution phases. The exhibit is based on statistical regression analysis, which determines how much of the outcome (wealth accumulation or sustainable withdrawal rate) can be explained by the returns experienced in each year of the life cycle. The exhibit isolates the impact of each year's return on lifetime outcomes using a larger sample of one hundred thousand Monte Carlo simulations based on a 50/50 portfolio with the same characteristics as the historical data. For the first thirty years (when individuals are saving), the percentage of the final wealth accumulation at the retirement date grows from year one through year thirty. With wealth accumulations at insignificant levels in the early part of one's career, the early returns have very little impact on the absolute level of wealth accumulated at the end of the savings period. But as retirement approaches, a given percentage return produces an increasing impact on the final wealth value in absolute terms, leaving individuals particularly vulnerable to these later returns. Simply put, later market returns impact more years of contributions.

In years thirty-one through sixty, during the retirement distribution phase, the exhibit shows the impact of each year's return on the maximum sustainable withdrawal rate. The return in year thirty-one represents the first year of retirement; this initial return explains almost 14 percent of the final outcome for retirees. Retirees are extremely vulnerable to what happens just after they retire. This result would hold even more so with the human capital considerations of the real world, as it is increasingly difficult to return to the workforce after you retire. Sustainable withdrawal rates are disproportionately explained by what happens in the early part of retirement. Returns from later in retirement have minimal impact, as the outcome for that retirement (high or low sustainable spending) was already set in motion earlier.

### *Exhibit 3.6*

#### *Lifetime Sequence-of-Returns Risk*

*50/50 Asset Allocation, Inflation-Adjusted Spending*

*100,000 Monte Carlo Simulations Based on S&P 500 Data, 1926–2016,*

*S&P 500 and Intermediate-Term Government Bonds*